	Pimpri Chinchwad Education Trust's **Pimpri Chinchwad College of Engineering & Research Ravet, Pune** **IQAC PCCOER**	
AY: 2025– 26	**Assignment No. 2**	Term- I

Date: 31.07.2025

Department: Computer Engineering **Class:** SE **Div:** E & F

Subject Name: Operating Systems

Course Objectives

To understand and analyze process management concepts.

Course Outcomes

CO No.	Course Outcomes	BT Level
P203.2	Apply process scheduling and synchronization to optimize CPU utilization in modern operating systems.	**BT4**

Sr. No.	Question	CO	PI
Q1	Define Process? List types of Process. Explain Process state transition diagram.	**P203.2** BT4	1.6.1, 2.5.3 3.5.2, 5.6.2 9.4.1, 12.4.1
Q2	Explain Process control block.	**P203.2** BT4	1.6.1, 2.5.3 3.5.2, 5.6.2 9.4.1, 12.4.1
Q3	Explain Context Switching and its impact on performance with the help of example.	**P203.2** BT4	1.6.1, 2.5.3 3.5.2, 5.6.2 9.4.1, 12.4.1
Q4	Solve following example using FCFS, SJF(Non-Preemptive) and Round Robin CPU Scheduling Algorithm. Time Quantum = 3 Process | Arrival Time | Burst Time P1 | 0 | 8 P2 | 5 | 2 P3 | 1 | 7 P4 | 6 | 3 P5 | 8 | 5	**P203.2** BT4	1.6.1, 2.5.3 3.5.2, 5.6.2 9.4.1, 12.4.1
Q5	Explain types of process schedulers with example.	**P203.2** BT4	1.6.1, 2.5.3 3.5.2, 5.6.2 9.4.1, 12.4.1

Q6	Solve following example using FCFS, SJF (Preemptive) Priority (Preemptive & Non-Preemptive).	**P203.2** BT4	1.6.1, 2.5.3 3.5.2, 5.6.2 9.4.1, 12.4.1

Process	Arrival Time	Burst Time	Priority
P1	2	2	3
P2	5	6	4
P3	0	4	1
P4	0	7	2
P5	7	4	5

Mrs. Swati K. Rajput
Mrs. Shrutika Menkudale
Mr. Pratik Chopade
Subject I/C